Lab: Defining Robotic Agents as Active Inference Systems

Objective

Design and analyze a robotic agent by specifying its generative model, belief states, and action repertoire. You will compare how classical control agents and Active Inference agents differ in their architecture, and explore how morphological computation and embodiment shape agency.

Prerequisites

- Understanding of what constitutes an agent (autonomy, goal-directedness, environmental coupling)
- Familiarity with the Markov blanket formalism from Module 01
- Basic understanding of Bayesian inference

Part 1: Agent Specification

Consider a warehouse robot that must navigate to pick-up locations, grasp objects, and deliver them to packing stations.

1. Define the agent's **generative model** by specifying:
2. Hidden states: robot pose (x, y, theta), gripper state (open/closed), object location, target station
3. Observations: LIDAR scans, camera images, force-torque readings at the gripper, wheel encoder counts

4. Actions: wheel velocities (v_left, v_right), gripper open/close command

5. What distinguishes this robot as an *agent* rather than merely a *system*? Identify at least three properties that confer agency.

Write your response here

Part 2: Classical vs. Active Inference Agent Architecture

Compare two designs for the warehouse robot:

Feature	Classical Agent	Active Inference Agent
State estimation	________________	________________
Goal representation	________________	________________
Action selection	________________	________________
Error handling	________________	________________
Adaptation mechanism	________________	________________

For the classical agent, assume a finite state machine with PID controllers. For the Active Inference agent, describe how beliefs, preferences (prior preferences over observations), and expected free energy drive behavior.

Write your response here

Part 3: Morphological Computation Analysis

Analyze how the robot's physical body contributes to its cognitive processing:

1. A compliant gripper passively conforms to object shapes without explicit shape modeling. How does this reduce the complexity of the generative model?
2. A round-bodied robot deflects off obstacles rather than requiring collision avoidance planning. What generative model states are eliminated?
3. Identify one additional example of morphological computation in robotics and describe its free energy implications.

Write your response here

Part 4: Belief Dynamics Trace

Trace the agent's belief updates through a scenario where it approaches a shelf but finds the expected object missing:

1. **Prior belief**: Object is at location (3.2, 1.5) on shelf B with high confidence.
2. **Observation**: Camera detects empty shelf at that location.
3. **Prediction error**: Describe the qualitative prediction error (expected vs. observed).
4. **Belief update**: How does the agent revise its beliefs? What happens to uncertainty?
5. **Action consequence**: What action does the agent select to resolve the remaining surprise?

Write your response here

Part 5: Autonomy Spectrum

Place different robotic systems on an autonomy spectrum based on their Active Inference properties:

Robot System	Generative Model Depth	Temporal Horizon	Adaptability
Thermostat			
Roomba vacuum			
Warehouse pick robot			
Surgical robot (da Vinci)			
Self-driving car			

Summary Table

Agent Property	Formal Definition	Warehouse Robot Implementation
Autonomy	Self-evidencing via free energy minimization	________________
Goal-directedness	Prior preferences over observations	________________
Adaptability	Model parameter updating	________________
Embodiment	Physical Markov blanket	________________

References

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